

A PROPOSAL FOR A SOFTWARE TRANSFORMATION PROJECT

AI-Powered Knowledge Search System — Companies House

Section 1: Business Case

1.1 Organisational Context

Version 1 is a technology services company contracted to Companies House (CH) — the UK registrar of companies and executive agency of the Department for Business and Trade. The Companies House Strategy 2020-2025 (Companies House, 2022) identifies digital transformation, data quality, and service efficiency as core strategic priorities — this proposal directly supports all three. The Version 1 delivery team of 20 professionals operates entirely within CH's own infrastructure: CH machines, CH VPN, and CH credentials. Despite this proximity, a systemic knowledge management failure has emerged that measurably undermines delivery performance and contradicts CH's strategic objectives.

1.2 The Problem

Research consistently identifies knowledge management as one of the primary productivity challenges in software development organisations (Souza et al., 2024; Kianto et al., 2019). McKinsey Global Institute (2012) estimates that knowledge workers spend approximately 19% of their working week searching for information — a figure Xia et al. (2017) place even higher for software engineers at 20-30%. At Companies House, this challenge is compounded by a microservice architecture whose institutional knowledge is fragmented across six disconnected platforms: Confluence, GitHub, Slack, Notion, Miro boards, and shared drives.

Stakeholder analysis and delivery records confirm the impact. At the start of each feature, the team raises a SPIKE — a formal timebox investigation — simply to understand existing system architecture. Delivery records show SPIKES of 3-5 days across all 8 features in the preceding 12 months, with at least 10 extending to a full week. Six of the 8 features required an additional sprint — a pattern Souza et al. (2024) identify as knowledge debt accumulation. The knowledge exists within permanent CH staff; the structural problem is the absence of a mechanism to surface it at the point of need.

1.3 Root Cause Analysis

1.3.1 SWOT Analysis

STRENGTHS	WEAKNESSES
Deep CH institutional knowledge exists. Experienced V1 team with 18+ months at CH. All documentation exists across six platforms.	No cross-platform search. Knowledge locked in Slack threads and Miro boards. No knowledge transfer process. High onboarding cost.
OPPORTUNITIES	THREATS
Mature RAG AI accessible to small teams. Open-source LLMs reduce cost. CH Strategy 2020-2025 explicitly supports digital transformation. Phase 2 extension to all CH contractors.	Evolving UK AI regulatory landscape (DSIT, 2023). Developer distrust if AI produces incorrect guidance. Government procurement may delay approval. Miro boards hard for AI to parse.

Table 1: SWOT Analysis — Version 1 Documentation Landscape at Companies House

1.3.2 Five Whys — Root Cause

- Developers spend 25% of every feature timeline on SPIKEs and discovery.
- WHY? Documentation fragmented across 6 platforms; mixed pipelines have no source of truth.
- WHY? Each platform and pipeline adopted independently with no knowledge architecture strategy.
- WHY? No designated owner of documentation infrastructure as CH microservice estate grew.
- ROOT CAUSE: Institutional knowledge has never been managed as a structured asset — creating accumulated knowledge debt consistent with the knowledge silo patterns identified by Souza et al. (2024) in software organisations, directly contradicting CH's 2020-2025 digital transformation strategy (Companies House, 2022).

1.3.4 Five Hows — Solution Pathway

Step	Question	Answer
How 1	How do we eliminate knowledge gaps?	Make all documentation searchable from one unified place
How 2	How do we make it searchable?	Index all 6 platforms into a unified RAG-based AI system
How 3	How do we make answers trustworthy?	Cite the source document with every AI-generated answer
How 4	How do we keep it current?	Automate re-indexing whenever source documents are updated
How 5	How do we prove it worked?	Track SPIKE duration in Jira before and after — measurement built into existing tooling

Table 2: Five Hows — Solution Pathway (ASQ, 2024)

1.4 Problem Statement

Version 1 developers contracted to Companies House experience significant and measurable delivery delays at the start of every feature, because critical technical and institutional knowledge — spanning microservice dependencies, deployment pipelines, and legacy system behaviour — is fragmented across disconnected platforms with no unified way to access it.

1.5 Solution Evaluation

Three options were evaluated using a weighted decision matrix (Kepner & Tregoe, 1965). Each rejected alternative is grounded in published research:

Criterion	Weight	Do Nothing	COTS (Glean/Guru)	RAG System
Cost to CH	30%	0/10 — waste continues	4/10 — per-seat licensing	9/10 — £12,100 only
UK GDPR compliance	25%	5/10 — no change	3/10 — data leaves CH	10/10 — within CH boundary
Solves root cause	25%	0/10 — persists indefinitely	6/10 — generic search only	10/10 — CH-specific knowledge
Speed of delivery	20%	10/10 — immediate	7/10 — fast but procurement	7/10 — 6-month build
WEIGHTED SCORE	100%	2.75	5.05	8.80 ✓

Table 4: Weighted Decision Matrix — Solution Evaluation (Kepner & Tregoe, 1965)

The RAG system scores 8.80, decisively ahead of all alternatives. Do nothing (2.75) contradicts CH's strategic objectives (Companies House, 2022). COTS (5.05) creates GDPR obligations through third-party data processing, requiring legal assessment that exceeds the project timeline (DSIT, 2023). Keyword federation fails because Xia et al. (2017) demonstrate developer information-seeking failure is a semantic mismatch problem — keyword matching cannot address it; vector search can.

1.6 Goal Statement

To eliminate the knowledge gap at the start of every feature delivery at Companies House by building a searchable AI system that surfaces microservice dependencies, pipeline configurations, and legacy system knowledge — turning weeks of investigation into minutes and recovering an estimated £160,000-£240,000 in annual delivery capacity, directly supporting the Companies House Strategy 2020-2025.

1.7 Impact Story and Cost-Benefit Analysis

CBA was selected over BIA because the primary justification is financial — converting productivity waste into monetary terms to support a government investment decision (Multiverse, 2026). BIA was evaluated but excels at dependency mapping rather than financial case-making.

Framework	Applied to This Project
PROBLEM	25% of every feature timeline consumed by SPIKEs, pipeline discovery, and dependency mapping — contradicting CH's 2020-2025 efficiency goals.
IMPACT	6/8 features last year needed extra sprints (£80,000 each). 10 week-long SPIKEs at £40,000 each. Total annual knowledge-driven waste: ~£592,000.
INVESTMENT	6-month project redirecting 70% of the existing V1 team (already contracted and paid by CH). True additional cost to CH: £12,100.
PAYOFF	Conservative 50% saving: £296,000/year. Payback on true additional cost: under 2 weeks. Phase 2 compounds returns further.

Table 5: Four-Part Impact Story

CBA Metric	Formula	Value (Conservative)
Annual extra sprint cost (6 features)	$6 \times 20 \times £400 \times 10 \text{ days}$	£480,000
Annual SPIKE cost (8x3-day + 10x5-day)	$(8 \times 20 \times £400 \times 3) + (10 \times 20 \times £400 \times 5)$	£592,000
True additional project cost	LLM API + governance + tooling + contingency	£12,100
Annual benefit (50% conservative recovery)	$£592,000 \times 50\%$	£296,000
Net Benefit Year 1	Annual Benefit - Additional Cost	£283,900
Benefit-Cost Ratio (BCR)	Benefits / Costs	24.5 : 1
ROI	$(\text{Net Benefit} / \text{Cost}) \times 100\%$	2,346%
Payback Period	Cost / Monthly Benefit	Under 2 weeks

Table 6: Cost-Benefit Analysis — Multiverse Module 5 Framework (Multiverse, 2026)

The projected BCR of 24.5:1 and ROI of 2,346% represent a compelling financial case based on conservative estimates. The existing team cost (~£873,600) is redirected CH contracted spend — the true incremental ask is £12,100, validated against Jira delivery data in the pilot phase.

Section 2: Business Analysis Methodology

The overarching framework is BABOK v3 (IIBA, 2015), covering Strategy Analysis, Elicitation, Requirements Analysis, and Solution Evaluation. Three analytical tools were selected and justified below.

2.1 PESTLE Analysis

PESTLE was selected as the primary environmental scanning tool because it specifically examines systemic external influences (Mutomorro, 2024) — essential for a government AI deployment. SWOT was evaluated as an alternative but rejected as the primary framework: its value depends entirely on research depth (BiteSize Learning, 2024) and it conflates internal/external factors less systematically than PESTLE (Johnson et al., 2017). SWOT was retained as a complementary synthesis tool in Section 1.3.1.

Factor	Key Finding
Political	CH is a government agency subject to GDS standards and Cabinet Office AI governance. CH Strategy 2020-2025 (Companies House, 2022) mandates digital transformation — this project directly supports that mandate.
Economic	Contractor rates of £400-600/day make wasted SPIKEs expensive. The £12,100 vs £296,000/year saving argument is compelling under public sector budget pressure.
Social	Developer scepticism towards AI without visible sources is documented (Stack Overflow, 2024). Mandatory citation directly addresses this barrier.
Technological	Open-source LLMs (Meta Llama 3) and vector databases (ChromaDB) make self-hosted RAG feasible at minimal cost. Mixed CH pipelines require Sprint 0 discovery.
Legal	UK GDPR Article 44-49 restricts data processing to UK jurisdiction. Operating entirely within CH infrastructure eliminates this risk.
Environmental	CH has government sustainability obligations. Llama 3.1 8B as default reduces energy consumption vs large commercial LLMs.

Table 7: PESTLE Analysis Applied to the AI Knowledge Search Project

2.2 Process Mapping and Value Stream Analysis

BPMN process mapping revealed an average of seven manual steps and four platform switches per developer query — a quantifiable baseline for post-implementation comparison. Value Stream Mapping (Poppendieck & Poppendieck, 2003) was applied alongside BPMN to identify non-value-adding activities. Use case modelling was evaluated as an alternative but reserved for the solution design phase, as it is less suited to current-state diagnosis across fragmented workflows.

2.3 MoSCoW Prioritisation and Iron Triangle

MoSCoW was selected to enforce scope discipline in an AI project where capability expansion pressure is constant (Clegg & Barker, 1994). The Iron Triangle (Atkinson, 1999; MindTools, 2024) frames the core constraint: Time fixed at 6 months, Cost fixed at £12,100 — any Scope increase reduces Quality. Kano Model was evaluated but requires extensive user research better suited to consumer products.

Category	Requirements
Must Have	RAG pipeline indexing all 6 CH platforms. Natural language query interface. Mandatory source citation. System within CH infrastructure. SPIKE measurement via Jira.
Should Have	Live SPIKE Reduction Dashboard on Confluence. Administrator dashboard. Answer quality feedback mechanism.
Could Have	Automated documentation freshness scoring. Knowledge gap detection. Extension to other CH contractor teams.
Won't Have	Code review, automated testing, deployment automation (Phase 2). LLM fine-tuning. Change management and staff upskilling.

Table 8: MoSCoW Requirements Prioritisation

2.4 Technology Component Justification

Each technical component is justified against evaluated alternatives (KSB S1 — Distinction criterion):

Component	Selected	Alternative Considered	Justification for Selection
Vector Database	ChromaDB (self-hosted)	Pinecone, Weaviate Cloud	Self-hosted within CH satisfies UK GDPR residency. No data leaves boundary. Zero additional licensing cost.
Primary LLM	GPT-4o via Azure OpenAI	Claude, Gemini, Llama 3	Azure already in CH procurement framework. UK data residency met. Hard spending cap configurable natively.
Fallback LLM	Llama 3.1 8B (open source)	GPT-3.5, Mistral	Zero API cost when activated. Sufficient for standard queries. Directly mitigates financial risk R5.
User Interface	Web app + Slack bot	Web app only	Slack bot removes context-switching — developers query within existing workflow, improving adoption.
Project Tooling	GitHub Projects + Jira	Asana, Microsoft Planner	GitHub integrates with CH codebase. Jira already tracks SPIKES — enables before/after measurement.

Table 9: Technology Component Justification

2.5 Project Management Methodology

Adapted Scrum was selected through systematic comparison. Stakeholder engagement revealed CH describes its approach as Agile but delivers legacy systems through sequential planning — a pattern documented as pseudo-Agile adoption (Dikert et al., 2016). Adapted Scrum acknowledges this reality and aligns with Peng et al.'s (2023) finding that AI tools deliver greatest benefit in iterative environments. Waterfall was rejected because AI technical unknowns cannot be planned upfront; full Hybrid because governance overhead is disproportionate for the project scale.

Criterion	Waterfall	Agile — Scrum	Hybrid
Technical unknowns	Low — plan obsolete on discovery	HIGH — sprints accommodate	Medium
Stakeholder feedback	Low — end of project only	HIGH — every 2 weeks	Medium
CH environment fit	Low — ignores hybrid reality	HIGH — adapted for CH context	Medium
VERDICT	Rejected	SELECTED	Rejected

Table 10: Methodology Comparison — Waterfall vs Agile vs Hybrid

Sprint 0 is a Design Sprint co-designing the architecture with pilot developers and DevOps. Sprint feedback is documented in Confluence, ensuring an auditable loop (Schwaber & Sutherland, 2020). GitHub Projects manages the Kanban board; SPIKE tickets continue in Jira for before/after measurement.

2.6 AI Ethics Framework

Responsible AI deployment requires ethical governance beyond legal compliance. This project adopts three pillars aligned to UK Government AI ethics principles (DSIT, 2023):

TRANSPARENCY	ACCOUNTABILITY	FAIRNESS
✓ Every answer cites its source ✓ Source platform shown with date ✓ No hidden reasoning or generation ✓ Developer sees exactly what was retrieved	✓ Technical Architects panel owns governance ✓ Human review encouraged for critical decisions ✓ Full audit trail maintained in Confluence ✓ Sponsor sign-off before production release	✓ System retrieves — does not generate independently ✓ No demographic data indexed or processed ✓ CH maintains document governance standards ✓ Sex and racial bias explicitly considered in design

Figure 1: AI Ethics Framework — Three Pillars Applied to the Knowledge Search System

Section 3: Project Charter

3.1 Project Overview

Project: AI-Powered Knowledge Search System | Technical Lead: [Version 1 Technical Lead] | Sponsor: CH Technical Architects Panel | PM: [V1 PM] | DM: [V1 DM] | Duration: 6 months | Start: [Month 1, 2026]

Goal: To eliminate the knowledge gap at the start of every feature delivery at Companies House by building a searchable AI system that surfaces microservice dependencies, pipeline configurations, and legacy system knowledge — turning weeks of investigation into minutes and recovering an estimated £160,000-£240,000 in annual delivery capacity, directly supporting the CH Strategy 2020-2025.

3.2 Capability Map

Capability	Current State	Future State
Knowledge Discovery	Manual, fragmented, 3-5 day SPIKES	Unified AI search, answers in minutes
Onboarding	Weeks of informal transfer, relying on senior staff	Day 1 self-service access to all CH knowledge
Pipeline Visibility	Scattered Concourse and Jenkins, no source of truth	All pipeline configs indexed and queryable
Legacy Knowledge	Lost when permanent CH staff unavailable	Permanently captured and accessible
Feature Delivery	6/8 features need extra sprint; 25% overhead	Features within original estimate; SPIKES eliminated

Table 11: Capability Map — Current vs Future State

3.3 SMART Deliverables

Deliverables were defined using the SMART framework — Specific, Measurable, Attainable, Relevant, and Time-based (Doran, 1981) — ensuring each outcome is objectively evaluable against the KPI matrix in Section 3.4:

ID	Deliverable	SMART Definition
D1	RAG Pipeline	Specific: RAG pipeline indexing all 6 CH platforms. Measurable: 80%+ recall on 50 benchmark queries. Attainable: standard RAG within CH. Relevant: eliminates SPIKE overhead. Time-based: Sprint 4 (Month 2).
D2	UI + Slack Bot	Specific: Web interface and Slack bot. Measurable: deployed to 10 pilot developers. Attainable: standard web and Slack API. Relevant: adoption in existing workflow. Time-based: Sprint 10 (Month 5).
D3	Governance Sign-Off	Specific: Technical Architects approval including AI ethics sign-off. Measurable: signed compliance document. Attainable: built within CH boundary. Relevant: required for production release. Time-based: Month 4.
D4	Pilot Report	Specific: Report covering SPIKE reduction, survey results, ROI and Phase 2 recommendations. Measurable: 40%+ SPIKE reduction vs Jira baseline. Attainable: 6-week pilot. Relevant: proves value. Time-based: Month 6.

Table 12: SMART Deliverables

3.4 KPI Matrix

The following KPIs provide measurable success criteria aligned to the SMART deliverables, enabling objective evaluation of project outcomes:

KPI	Baseline	Target	Measurement	Frequency
Average SPIKE duration	3-5 days	Under 1 day	Jira ticket duration	Per sprint
Features requiring extra sprint	6/8 (75%)	0/8 (0%)	Sprint delivery records	Per feature
Developer query resolution time	30+ minutes	Under 2 minutes	Developer survey + system logs	Monthly
Annual delivery waste recovered	~£592,000	£296,000+ (50% min)	Jira + sprint records	6-monthly
System recall accuracy	N/A	80%+ benchmark	50-query benchmark test	Pre-launch + monthly
Developer trust score	N/A	4.0/5.0+	Anonymous pilot survey	M3 and M5

Table 13: KPI Matrix — Success Measurement Framework

3.5 Scoping Statement

In Scope:

- RAG pipeline indexing Confluence, GitHub, Slack, Notion, Miro, and shared drives
- Natural language query interface and Slack bot within CH infrastructure
- Mandatory source citation and AI ethics governance sign-off
- SPIKE duration measurement using existing Jira reports and KPI tracking
- User training materials and system documentation

Out of Scope (this phase):

- Code review and automated testing (Phase 2)
- LLM fine-tuning, platform replacement, change management, production database integration

3.6 Work Breakdown Structure

Workstream	Key Tasks	Owner	Sprints
WS1: Sprint 0 + Infrastructure	Co-design workshop; provision within CH; LLM config; CI/CD; sprint backlog; KPI baseline; ethics governance	Astha + DevOps	S1-S2
WS2: Data Ingestion	Connectors for all 6 platforms; chunking and embedding; Miro audit and Confluence conversion; boundary validation	Senior Devs	S2-S4
WS3: RAG Core Engine	Retrieval logic; prompt engineering; citation layer; hallucination benchmark (50 queries); performance optimisation	Astha + Devs	S4-S7
WS4: UI + Governance	Web interface; Slack bot; UX testing; governance and AI ethics sign-off; Live SPIKE Dashboard on Confluence	Devs + BA	S7-S10
WS5: Pilot + Evaluation	10-developer pilot; Jira KPI tracking; developer survey; ROI calculation; final report and Phase 2 plan	PM + QA + Astha	S10-S12

Table 14: Work Breakdown Structure

3.7 Sprint Chronogram

Workstream	S1	S2	S3	S4	S5	S6	S7	S8	S9	S10	S11	S12
WS1: Sprint 0												
WS2: Data Ingestion												
WS3: RAG Engine												
WS4: UI + Governance												
WS5: Pilot												
Month	1	1	2	2	3	3	4	4	5	5	6	6

Table 15: Sprint Chronogram — 12 x Two-Week Sprints

3.8 Budget

Cost Item	Category	Planned	Actual	Variance
Existing V1 team 70% x 6 months (contracted rates — available on request)	Redirected Labour	~£873,600	[TBC]	[TBC]
LLM API — £800/month hard cap x 6 months	Technology	£4,800	[TBC]	[TBC]
Additional tooling and licences	Technology	£1,200	[TBC]	[TBC]
Governance and AI ethics review	Professional Services	£3,000	[TBC]	[TBC]
Risk mitigation reserve (Section 4)	Risk Reserve	£2,000	[TBC]	[TBC]
Contingency buffer (10%)	Contingency	£1,100	[TBC]	[TBC]
TRUE ADDITIONAL COST TO CH		£12,100	[TBC]	[TBC]
TOTAL (incl. redirected team)		~£885,700	[TBC]	[TBC]

Table 16: Project Budget with Variance Tracking

Variance = Planned - Actual. Positive = underspend; negative triggers sponsor review. Redirected team cost is existing CH contracted spend. Contingency follows Multiverse guidance (2026).

Section 4: Risk Management Plan

Risk management follows ISO 31000:2018 and PMI PMBOK (PMI, 2021). Risks were identified through SWOT, PESTLE, stakeholder interviews, and technical discovery. Exposure Rating = Probability x Impact (Low=0.1, Medium=0.5, High=0.9): above 0.25 = HIGH; 0.10-0.24 = MEDIUM; below 0.10 = LOW.

4.1 Risk Register

ID	Type	Risk	Prob.	Impact	Score	Rating	Mitigation + Response	Owner
R1	Compliance	SC-cleared data accidentally indexed	Low	High	0.09	LOW	AVOIDANCE: Explicitly exclude SC-cleared repos from indexing config at Sprint 0. Data flow diagrams reviewed by CH security.	Astha
R2	Operational	Permanent CH staff resist knowledge sharing	Medium	Medium	0.25	HIGH	REDUCTION: Frame as amplifying expertise not replacing it. Involve 2 CH staff in Sprint 0. Cost: £200.	Astha
R3	Strategic	ROE delivery slows during 70% redirect	High	Medium	0.45	HIGH	REDUCTION: Written agreement with Technical Architects on reduced ROE targets before project start. Monthly updates.	DM
R4	Operational	Pipeline complexity makes indexing difficult	High	Medium	0.45	HIGH	REDUCTION: Sprint 0 maps all Concourse and Jenkins locations before build. DevOps in co-design workshop. Cost: £200.	Astha
R5	Reputational	LLM hallucination erodes developer trust	Medium	High	0.45	HIGH	REDUCTION: LLM constrained to retrieved context only. Mandatory source citation. 50-query benchmark pre-launch. Cost: £300.	Astha
R6a	Strategic	Technical Architects approval delayed	Low	Low	0.01	LOW	REDUCTION: Leverage existing ROE relationship for informal pre-approval.	Astha
R6b	Strategic	Government financial procurement delays start	Medium	High	0.45	HIGH	REDUCTION: Financial approval submitted 8 weeks before planned start alongside Technical Architects.	DM
R7	Operational	Key person dependency	Low	High	0.09	LOW	TRANSFERENCE: All decisions	Astha

		on Technical Lead					documented in Confluence. IaC. Senior developer named as deputy. Cost: £200.	
R8	Operational	Miro boards not machine readable	High	High	0.81	HIGH	REDUCTION: Sprint 0 Miro audit — critical diagrams converted to Confluence text. BA establishes documentation guidelines. Cost: £500.	BA Team
R9	Reputational	V1 contract renewal risk if ROE slows	Low	High	0.09	LOW	REDUCTION: Pre-project written agreement on ROE targets. Strong existing relationship mitigates.	DM+Astha

Table 17: Risk Register — ISO 31000:2018 and PMI PMBOK Framework

4.2 Risk Heat Map

Likelihood →	LOW Impact	MEDIUM Impact	HIGH Impact
HIGH		R3, R4	R8
MEDIUM		R2	R5, R6b
LOW	R6a		R1, R7, R9

Table 18: Risk Heat Map

4.3 Opportunities Register

ID	Opportunity	Likelihood	Impact	How to Realise
O1	Phase 2 — code review and automated testing	High	High	Design for extensibility from day one. Phase 2 proposal at evaluation.
O2	System replicated across all CH contractor teams	High	High	Document learnings. Include scalability in architecture from day one.
O3	Version 1 wins additional CH work through demonstrated AI innovation	High	High	Delivery Director briefed. CH reference case prepared for V1 marketing.

Table 19: Opportunities Register (PMI, 2021 — KSB K3)

4.4 Monitoring

The risk register is reviewed fortnightly at every sprint retrospective. HIGH escalations trigger written sponsor notification within 48 hours. A formal mid-project review is presented at Month 3. Risk mitigation costs of £1,400 are incorporated into the budget. The evolving UK AI regulatory landscape (DSIT, 2023) is specifically monitored — binding obligations anticipated by 2027 may require architectural adjustments.

Section 5: Communication Plan

The communication plan follows the three-step stakeholder analysis process — Identification, Categorisation, Engagement (Multiverse, 2026) — and applies the Minto Pyramid and Elaboration Likelihood Model to tailor messaging by audience.

5.1 Stakeholder Identification and Categorisation

Eight stakeholder groups were identified using a mindmap approach (Visual Paradigm, 2024) — asking who is affected, who decides, who funds, and who must approve. Categorisation applied two complementary models: Mendelow's Matrix (Mendelow, 1991; Oxford College of Procurement and Supply, 2024) on Power and Interest, and the Salience Model (Mitchell et al., 1997; projectmanagers.net, 2024) adding Urgency. The Salience Model captures time-sensitivity that Mendelow's two-dimensional grid cannot — particularly critical given government procurement timeline risk (R6b).

Stakeholder	Power	Interest	Urgency	Mendelow	Strategy
Technical Architects Panel	High	High	High	Manage Closely	Monthly formal update + open sprint reviews. Logos-led.
CH Finance and Procurement	High	Low	Med	Keep Satisfied	Submit documentation 8 weeks early. Ethos + Logos.
CH Security and Governance	High	Low	Low	Keep Satisfied	Proactive data flow diagrams at Sprint 0. Sign-off at M4.
Permanent CH Employees	Med	Med	Low	Keep Informed	Weekly Slack. Involve in Sprint 0. Pathos — expertise amplified.
V1 Development Team	Low	High	High	Keep Informed	Daily standups, sprint demos, retrospectives. Pathos-led.
CH End Users	Low	High	Low	Keep Informed	Monthly roadmap updates. Features on time, priorities respected.
V1 Delivery Director	High	Med	Low	Keep Satisfied	Monthly headline. Logos — contract value and Phase 2.

Table 20: Stakeholder Categorisation — Mendelow and Salience Model Combined

5.3 Engagement — RACI Matrix

Deliverable	Astha	V1 Team	Tech Arch	Finance	CH Staff	V1 Dir.	Security	End Users
Project Approval	C	I	A	C	I	C	I	I
D1: RAG Pipeline	R/A	R	I	I	C	I	I	I
D2: UI + Slack	R/A	R	C	I	C	I	I	I
D3: Governance	R/A	I	A	I	I	I	C	I
D4: Pilot Report	R/A	C	A	C	I	I	I	I

Table 21: RACI Matrix — R=Responsible, A=Accountable, C=Consulted, I=Informed

5.4 Communication Cadence

Communication	Audience	Vehicle	Frequency	Purpose and Approach
Formal Project Update	Technical Architects	Presentation + report	Monthly	Logos: SPIKE KPI data, ROI, sprint velocity, AI ethics compliance.
Sprint Review + Demo	All CH (open)	Live demo	Fortnightly	Live system answers real developer questions. Drives adoption.
Slack Updates	Permanent CH Staff	Existing channel	Weekly	Pathos + Ethos: celebrate contributions. Expertise valued.
Delivery Director Update	V1 Director	Email + call	Monthly	Logos: contract status, Phase 2 pipeline.
Feature Roadmap	CH End Users	Confluence	Monthly	Pathos: priorities confirmed, delivery on track.
Finance Submission	CH Finance	Formal pack	8 weeks pre-start	Ethos + Logos: governance evidence, BCR 24.5:1.
Live SPIKE Dashboard	Technical Architects	Confluence live page	Real-time	Innovative: live Jira-fed KPIs — SPIKE before vs after — builds confidence between formal meetings.

Table 22: Communication Cadence

5.5 Minto Pyramid and Elaboration Likelihood Model

The Minto Pyramid (Minto, 1996) leads with the main conclusion supported by three evidence-backed arguments. The Elaboration Likelihood Model (Petty & Cacioppo, 1986) predicts that high-motivation audiences process via the central route (Logos), whilst lower-motivation audiences respond to peripheral cues (Ethos and Pathos).

The pyramid below illustrates the communication structure for the CH Technical Architects Panel — the highest-priority 'Manage Closely' stakeholder:

MAIN MESSAGE				
Approve £12,100 to save £296,000 annually <i>directly advancing the CH Strategy 2020-2025</i>				
ARG 1 Problem proven		ARG 2 Solution low risk		ARG 3 Team proven
6/8 features — extra sprints	Jira SPIKE data	CH infra — no data leaves	AI ethics governed	Prior delivery proven

Figure 3: Minto Pyramid — Communication Structure for CH Technical Architects Panel

The pyramid is adapted for each stakeholder group:

Stakeholder	ELM Route	Appeal	Minto: Main Point + 3 Supporting Arguments
Technical Architects Panel	Central	Logos	MAIN: Approve £12,100 to save £296,000, advancing CH 2020-2025 strategy. ARG1: Jira data — 6/8 features with extra sprints. ARG2: Zero data risk — within CH infrastructure. ARG3: Proven team — prior transformation delivered.
CH Finance	Central	Logos + Ethos	MAIN: Fully governed, exceptional value. ARG1: BCR 24.5:1, ROI 2,346%. ARG2: No data leaves CH boundary. ARG3: Technical Architects endorsed.
Permanent CH Staff	Peripheral	Pathos + Ethos	MAIN: Your knowledge gets the visibility it deserves. ARG1: Expertise amplified not replaced. ARG2: New joiners self-serve — fewer interruptions. ARG3: Contributions credited in every answer.
V1 Delivery Director	Central	Logos	MAIN: Strengthens CH contract and creates Phase 2 opportunity. ARG1: AI innovation capability proven. ARG2: Phase 2 expands contract value. ARG3: CH becomes reference case.
CH End Users	Peripheral	Pathos	MAIN: Your features delivered on time. ARG1: No unexpected knowledge delays. ARG2: Priorities drive the roadmap. ARG3: Same trusted team, faster delivery.

Table 24: Minto Pyramid and ELM Applied by Stakeholder

Section 6: Justification Statement

6.1 Personal Drivers and Motivation

This project originates from direct observation of a recurring delivery pattern at Companies House, validated by research and stakeholder evidence. The CH Strategy 2020-2025 (Companies House, 2022) commits to efficient digital service delivery, yet delivery records show 25% of each feature's timeline consumed by knowledge discovery overhead. Souza et al. (2024) and McKinsey Global Institute (2012) confirm this is systemic in software organisations — not unique to CH. The Technical Lead role demands identification of structural barriers to performance. This proposal addresses that responsibility whilst simultaneously creating a commercial opportunity for Version 1 to demonstrate proactive innovation rather than reactive delivery.

6.2 Key Decisions and Their Rationale

RAG was selected over model fine-tuning on three research-backed grounds (Gao et al., 2023): fine-tuning requires significant GPU compute cost; hallucination risk increases because parametric memory can override retrieved facts; and periodic retraining is required as documentation evolves. RAG updates in near-real-time, grounds every answer in retrieved evidence, and requires no retraining — making it more reliable, maintainable, and cost-effective for a living knowledge base. Peng et al. (2023) further demonstrate that AI developer tools deliver greatest value when grounded in specific organisational context — a retrieval-based architecture is better suited to this requirement than a fine-tuned general model.

Running within CH's infrastructure was a governance decision — external architecture triggers GDPR assessment exceeding the timeline (DSIT, 2023). The 70/30 team split was selected over pausing ROE (contract risk) and using an external team (onboarding overhead). Adapted Scrum over Waterfall was selected because AI unknowns cannot be planned upfront — iterative sprints surface constraints early (Dikert et al., 2016).

6.3 Approaches Considered and Excluded

Three alternatives were evaluated through the weighted decision matrix (Table 4), each rejection grounded in published research. COTS (5.05 vs RAG's 8.80) was rejected on governance grounds — third-party data processing triggers GDPR assessment inconsistent with UK public sector AI guidance (DSIT, 2023). Keyword federation was rejected because Xia et al. (2017) demonstrate developer information-seeking failure is a semantic mismatch problem — vector search addresses this; keyword matching cannot. A knowledge graph approach (Hogan et al., 2021) was evaluated as a contrarian alternative — powerful for structured domains but requiring manual curation poorly suited to Slack threads, Confluence pages, and Miro boards.

6.4 AI Ethics Reflection

Responsible AI deployment requires ethical governance beyond legal compliance. The three-pillar framework (Figure 1) addresses the specific risks of a government AI system. Transparency is achieved through mandatory source citation — developer trust depends critically on answer traceability (Peng et al., 2023). Accountability is assigned to the CH Technical Architects panel; human review is recommended for critical decisions. On fairness, sex and racial bias were explicitly considered — the system retrieves and cites rather than generating independently, so content quality is governed by CH's own documentation standards. This aligns with UK Government responsible AI principles (DSIT, 2023).

6.5 Reflection on Approach

The communication strategy applies the ELM and Minto Pyramid (Figure 3) to calibrate persuasion to each stakeholder's motivation rather than broadcasting a single message (Petty & Cacioppo, 1986). The KPI matrix ensures success is evaluated objectively against pre-defined baselines. The proposal builds on prior successful delivery of a complex transformation within CH — evidence of the team's capacity to execute at scale. The system is more tightly scoped, carries a stronger financial case, and is governed by an explicit AI ethics framework — conditions that collectively support a robust case for approval.

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